

INPUT:

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#P1
print("Hello World")

#P2
usertext = input("What is your name? ")
print("Hello", usertext)

#P3
num1 = input('Enter first number: ')
num2 = input('Enter second number: ')

sum = float(num1) + float(num2)
print('The sum of {0} and {1} is {2}'.format(num1, num2, sum))

#P4
num1 = input('Enter first number: ')
num2 = input('Enter second number: ')

average =(int(num1) + int(num2))

print('average:{0} '.format(average))

#P5
visagrade = input('enter your visa grade : ')
finalgrade = input('enter your final grade : ')
average =(float(visagrade)*0.3)+(float(finalgrade)*0.7)
print("average :{0} ".format(average))

#P6
firstexam = input('your first exam : ')
secondexam = input('your second exam : ')
thirdexam = input('your third exam : ')
average =(float(firstexam)+float(secondexam)+float(thirdexam))/3
print("average :{0} ".format(average))

#P7
average = input('enter average : ')
if(int(average)>=50):
    print("Passed")
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else:
    print("Failed")

#P8
num = int(input("Enter a number: "))
if (num % 2) == 0:
    print("{0} is Even".format(num))
else:
    print("{0} is Odd".format(num))

#P9
num = float(input("Enter a number: "))
if num > 0:
    print("Positive number")
elif num == 0:
    print("Zero")
else:
    print("Negative number")

#P10
print("body mass index calculation program")
height = float(input("enter height (m):"))
weight = int(input("enter weight (kg):"))

index = weight/(height*height)

if index <=18:
    print("\n underweight BMI:{} ".format(index))
elif index > 18 and index <=25 :
    print("\n overweight BMI:{} ".format(index))
elif index > 25 and index <=30:
    print("\n obese BMI:{} ".format(index))
elif index > 30:
    print("\n severely obese BMI:{} ".format(index))

#P11
age = input('enter age : ')
if(int(age)<18):
    print("Your Age Is Not Eligible To Get A Driver's License")
else:

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print("Your Age Is Eligible To Get Your License")
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#P12
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```
for i in range(1,101):  
    print(i)
```

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#P13
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```
for i in range(1,101):  
    if i%2==0:  
        print(i)
```

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#P14
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```
for i in range(1,101):  
    if i%2!=0:  
        print(i)
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#P15
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```
for i in range(1,101):  
    if i%3==0 or i%5==0:  
        print(i)
```

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#P16
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```
num = input('enter number : ')  
for i in range(1,int(num)+1):  
    print(i)
```

```
#P17
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```
short = input('Enter short side : ')  
tall = input('Enter tall side : ')  
area = int(short) * int(tall)  
perimeter = 2 * (int(short) + int(tall))  
print("area: {0}".format(area))  
print("perimeter: {0}".format(perimeter))
```

```
#P18
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```
word = 'mrhuseyin'  
for char in word:  
    print(char)
```

```
#P19
```

```
sumofnumbers = 0
num1 = input('first number: ')
num2 = input('second number: ')
for i in range(int(num1) + 1, int(num2)):
    sumofnumbers += i
print("Sum of numbers between {0} and {1} : {2}".format(num1, num2,
sumofnumbers))
```

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#P20
```

```
selection = input("Press (1) for Cinema, (2) for Theater : ")
student = input("Are you student(Y/N) : ")
price = 0
#non-discounted fee calculation
if selection == '1':
    price = 10 #cinema
elif selection == '2':
    price = 5 #theatre
#student discount
if student == 'Y' or student == 'y':
    price = price / 2 #50
    print(" The fee you have to pay : {}".format(price))
```

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#P21
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```
num = int(input("Enter a number: "))
if num > 1:
    for i in range(2, num):
        if (num % i) == 0:
            print(num, "is not a prime number")
            print(i, "times", num // i, "is", num)
            break
        else:
            print(num, "is a prime number")
    else:
        print(num, "is not a prime number")
```

```
#P22
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```
NumList = []
Even_Sum = 0
Odd_Sum = 0
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Number = int(input("Please enter the Total Number of List Elements: "))
for i in range(1, Number + 1):
    value = int(input("Please enter the Value of %d Element : " %i))
    NumList.append(value)
for j in range(Number):
    if(NumList[j] % 2 == 0):
        Even_Sum = Even_Sum + NumList[j]
    else:
        Odd_Sum = Odd_Sum + NumList[j]
print("\nThe Sum of Even Numbers in this List = ", Even_Sum)
print("The Sum of Odd Numbers in this List = ", Odd_Sum)

```

#P23

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newsalary = 0
salary = input("enter new salary : ")
raiserate = input("salary raise rate(%) : ")
newsalary = int(salary)+(int(salary)*int(raiserate)/100)
print("increased salary :",newsalary)

```

#P24

```

import math
def find_Diameter(radius):
    return 2 * radius
def find_Circumference(radius):
    return 2 * math.pi * radius
def find_Area(radius):
    return math.pi * radius * radius
r = float(input(' Please Enter the radius of a circle: '))
diameter = find_Diameter(r)
circumference = find_Circumference(r)
area = find_Area(r)
print("\n Diameter Of a Circle = %.2f" %diameter)
print(" Circumference Of a Circle = %.2f" %circumference)
print(" Area Of a Circle = %.2f" %area)

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#P25

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def areaRectangle(a, b):
    return (a * b)
def perimeterRectangle(a, b):

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        return (2 * (a + b))
a = 5;
b = 6; print ("Area = ", areaRectangle(a, b))
print ("Perimeter = ", perimeterRectangle(a, b))

#P26
import random
import math

# Taking Inputs
lower = int(input("Enter Lower bound:- "))
upper = int(input("Enter Upper bound:- "))
# generating random number between the lower and upper
def new_func(lower, upper):
    return random.randint(lower, upper)

x = new_func(lower, upper)

# Calculate and save total allowed chances
total_chances = round(math.log(upper - lower + 1, 2))
print("\n\tYou've only ", total_chances, " chances to guess the
integer!\n")

# Initializing the number of guesses.
count = 0

# for calculation of minimum number of guesses depends upon range
while count < total_chances:

    count += 1
# taking guessing number as input
    guess = int(input("Guess a number:- "))
# Condition testing
    if x == guess:
        print("Congratulations you did it in ", count, " try")
        break
    elif x > guess:
        print("You guessed too small!")
    elif x < guess:
        print("You Guessed too high!")

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# If Guessing is more than required guesses,
# shows this output.
if count >= math.log(upper - lower + 1, 2):
    print("\nThe number is %d" % x)
print("\tBetter Luck Next time!")

#P27
import datetime
date = input('Enter the date(for example:09 02 2019):')
day_name = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday',
'Saturday', 'Sunday']
day = datetime.datetime.strptime(date, '%d %m %Y').weekday()
print(day_name[day])

#P28
def find_missing(lst):
    return [x for x in range(lst[0], lst[-1]+1)
if x not in lst]
# Driver code
lst = [1, 2, 4, 6, 7, 9, 10]
print(find_missing(lst))
[3, 5, 8]

#P29
char_list = ["a", "b" ,"c"]
string = "abcd"
matched_list = [characters in char_list for characters in string]
print(matched_list)
[True, True, True, False]
string_contains_chars = all(matched_list)
print(string_contains_chars)

#P30
total = 0
evenSums = 0
oddSums = 0
evenCount = 0
oddCount = 0

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done = False

while not done:
    user_in = input("Give me an integer or type 'done' to be done.")
    if user_in.lower() == "done":
        done = True
    else:
        num = int(user_in)
        total += num
        if num % 2 == 0:
            evenSums += num
            evenCount += 1
        else:
            oddSums += num
            oddCount += 1

evenAverage = evenSums / evenCount if evenCount > 0 else 0
oddAverage = oddSums / oddCount if oddCount > 0 else 0

print(total)
print("Even Average: " + str(evenAverage))
print("Odd Average: " + str(oddAverage))

```

OUTPUT:

```

Hello World
What is your name? ryan
Hello ryan
Enter first number: 1
Enter second number: 2
The sum of 1 and 2 is 3.0
Enter first number: 3
Enter second number: 4
average:7
enter your visa grade : 50
enter your final grade : 70
average :64.0
your first exam : 23
your second exam : 24
your third exam : 25
average :24.0
enter average : 26
Failed

```

Enter a number: 37

37 is Odd

Enter a number: 43

Positive number

body mass index calculation program

enter height (m):169

enter weight (kg):97

underweight BMI:0.0033962396274640244

enter age : 23

Your Age Is Eligible To Get Your License

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enter number : 24

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23

24

Enter short side : 20

Enter tall side : 25

area: 500

perimeter: 90

m

r

h

u

s

e

y

i

n

first number: 1

second number: 2

Sum of numbers between 1 and 2 : 0

Press (1) for Cinema, (2) for Theater : 1

Are you student(Y/N) : y

The fee you have to pay :5.0

Enter a number: 3

3 is a prime number

3 is not a prime number

Please enter the Total Number of List Elements: 4

Please enter the Value of 1 Element : 1

Please enter the Value of 2 Element : 2

Please enter the Value of 3 Element : 3

Please enter the Value of 4 Element : 4

The Sum of Even Numbers in this List = 6

The Sum of Odd Numbers in this List = 4

enter new salary : 50

salary raise rate(%) : 15

increased salary : 57.5

Please Enter the radius of a circle: 90

Diameter Of a Circle = 180.00

Circumference Of a Circle = 565.49

Area Of a Circle = 25446.90

Area = 30

Perimeter = 22

Enter Lower bound:- 50

Enter Upper bound:- 75

You've only 5 chances to guess the integer!

Guess a number:- 5

You guessed too small!

Guess a number:- 9

You guessed too small!

Guess a number:- 11

You guessed too small!

Guess a number:- 15

You guessed too small!

Guess a number:- 50

You guessed too small!

The number is 63

Better Luck Next time!

Enter the date(for example:09 02 2019):30 08 2007

Thursday

[3, 5, 8]

[True, True, True, False]

False

Give me an integer or type 'done' to be done.done

0

Even Average: 0

Odd Average: 0

